Supplementary Material

LLM Prompt Templates

for

"TextBraTS: Bridging 3D MRI, Text, and Genotype for Data-Efficient Glioma Segmentation and Analysis"

Overview

The TextBraTS text generation pipeline consists of two stages:

1. **Stage 1 — Image-to-Text Generation (GPT-4o):** Multi-parametric MRI video frames (FLAIR + T1CE with lesion overlays) are fed to GPT-4o to produce structured radiology-style reports describing lesion localization, edema extent, necrosis characteristics, and mass effect.
2. **Stage 2 — Semantic Distillation (DeepSeek-R1-8B):** The GPT-4o reports are distilled into a compact 4-section structured format via a locally deployed DeepSeek-R1-8B model (Ollama). A quality-control pipeline (`LLM_check.py`) further verifies syntactic and semantic correctness with automatic retries.

All prompts are reproduced verbatim below.

1 Stage 1: GPT-4o — Image-to-Text Generation

Model Configuration

- **Model:** GPT-4o (`gpt-4o`, OpenAI API)
- **Decoding:** $max_tokens = 1000$, temperature at default
- **Input:** Two video files (FLAIR modality + T1CE with lesion annotation overlays)
- **Output:** Structured radiology summary report

System Prompt

Listing 1: GPT-4o system prompt for radiology report generation

```
You are an advanced AI assistant specialized in analyzing brain MRI images and generating detailed radiological summary reports. Below are the tasks you need to perform and the specific requirements for report generation:

Input Data:

Two video files: one is a brain MRI in Flair modality, and the other is a corresponding annotated video showing lesion regions.

Each frame contains lesion annotations and signal intensity information.

Output Data:

Generate a detailed radiological summary report, including descriptions of the overall lesion region, edema region, necrotic region, and midline/ventricular compression.

Analysis Steps:

Identify brain regions:

Frontal Lobe

Temporal Lobe

Parietal Lobe

Occipital Lobe

Analyze the images:

Extract the annotated lesion region from each frame.

Calculate and classify signal intensity: high, isointense, or low.

Determine which brain region the lesion belongs to.

Generate the radiological summary report:

```

```
Overall lesion region: Describe the main brain regions involved. Indicate if lesions show mixed signal intensities (high and low) and/or punctate high-signal areas, suggesting multifocal lesions.

Edema region: Identify areas with obvious edema signals and indicate the affected regions. Describe the extent of tissue swelling and edema size.

Necrotic region: Describe regions potentially containing necrosis, including signal intensity and heterogeneity. Indicate whether necrosis is localized or scattered.

Midline and ventricular compression: Describe any midline shift or ventricular compression. Indicate whether the lesion exerts pressure on brain structures or ventricles.

Please use the above information and steps to batch-process the input brain MRI data and generate the corresponding radiological summary reports. Only the summary report section is required.
```

2 Stage 2: DeepSeek-R1-8B — Semantic Distillation

Model Configuration

- **Model:** DeepSeek-R1-8B (`deepseek-r1:8b`, Ollama)
- **Deployment:** Local, subprocess CLI (`ollama run deepseek-r1:8b`)
- **Decoding:** Temperature at default; *timeout* = 120–180 s
- **Target Format:** `Lesion: [Loc] | Edema: [Extent] | Necrosis: [Status]; [Type] | Ventricles compression: [Status]`
- **Postprocessing:** Strip `<think>...</think>` blocks, normalize separators, validate 4-section structure, filter forbidden words (`lobe`, `bilateral`, `hyperintense`, etc.)

2.1 2a. Main Distillation Prompt

This is the primary prompt used in `distill_with_ollama.py` to convert raw GPT-4o radiology reports into the structured 4-section format.

Listing 2: DeepSeek-R1-8B: Main semantic distillation prompt (`distill_with_ollama.py`)

```
You are an expert radiologist. Task: "Semantic Distillation" of MRI reports.

GOAL: Extract specific clinical attributes into a STRICT format.

OUTPUT FORMAT STRING:
"Lesion: [Loc] | Edema: [Extent] | Necrosis: [Status]; [Type] | Ventricles compression: [Status]"

STRICT EXTRACTION RULES:

1. Lesion (1st) - LOCATION ONLY:
   - **Content**: Extract Anatomical Region ONLY.
   - **Structure**: "[Side] [Region]". (Space between Side and Region).
   - **Separators**: Use **COMMA** (‘,‘) to separate different regions.
   - **Constraints**:
     - **OMIT "lobe"**: Output "Left frontal", NOT "Left frontal lobe".
     - **NO Slashes between Side/Region**: "Left frontal" (Correct) vs "Left/frontal " (WRONG).
     - **Split Bilateral**: "Left/Right frontal" (Same region) OR "Left frontal, Right parietal" (Diff regions).
     - **NO Signal/Shape**: Do not mention intensity or shape.

2. Edema (2nd) - EXTENT ONLY:
   - **DO NOT** extract location.
   - Extent:
     - "Extensive" (significant/spreading).
     - "Focal" (mild/local/surrounding/present).
     - "None" (explicitly absent).
   - DO NOT mention signal.

3. Necrosis (3rd) - DISTRIBUTION:
```

```
   - Status: "Present" or "Absent".
   - Type: "Heterogeneous" (mixed/scattered), "Concentrated" (core/focal), or "None
     ↪ ".
   - DO NOT mention signal.

4. Ventricles compression (4th) - BINARY:
   - Output ONLY: "Present" or "Absent".

5. General:
   - Handle Negation: "not observed" -> "Absent" or "None".
   - NO <think> tags. NO markdown.
```

2.2 2b. MNI Coordinates to Text Prompt

This prompt is used in `coords_to_text.py` for an alternative pathway: converting MNI spatial annotations of the BraTS-2021 non-overlap sample into the same structured format, enabling external validation.

Listing 3: DeepSeek-R1-8B: MNI-coordinate-to-text prompt (`coords_to_text.py`)

```
You are an expert neuroradiologist. Given MNI spatial annotations of brain tumor, produce a distilled clinical text report.

OUTPUT FORMAT (one line only):
Lesion: [Loc] | Edema: [Extent] | Necrosis: [Status]; [Type] | Ventricles compression: [Status]

RULES:

MNICONVENTIONS: x<0=Left x>0=Right. y>10=frontal, y=[-30,10]=temporal, y=[-55,-30]=parietal, y<-55=occipital.

1. LESION: Determine laterality from MNI x-coordinates: x<0=Left x>0=Right. Describe  ALL regions from all_regions.
- Bilateral same region: Left/Right frontal
- Different regions: Left frontal, Right parietal
- Omit lobe word. NO signal/shape words.

2. EDEMA: WT_vol>35000 -> Extensive. WT_vol<8000 -> Focal. Default Extensive.

3. NECROSIS: TC_vol>100 -> Present, else Absent. TC/(TC+ET)>0.4 -> Concentrated, else Heterogeneous. Absent -> Type=None.

4. VENTRICLES: Use ventricle_compression field: True->Present, False->Absent.

EXAMPLE:
Input: all_regions=[frontal,temporal] | TC_mni=(-28,18,42) WT_mni=(-26,14,38) ET_mni=(-28,17,40) | TC=5000 ET=9000 WT=30000 | vc=False
Output: Lesion: Left frontal, Left temporal | Edema: Extensive | Necrosis: Present; Heterogeneous | Ventricles compression: Absent

Output ONLY the single line with | separators. NO markdown. NO thinking. NO Chinese.
```

2.3 2c. Quality Control — Generator Prompt

When the 5-step state machine in `LLM_check.py` detects syntactic or semantic errors in the distilled output, it invokes the following few-shot regeneration prompt.

Listing 4: DeepSeek-R1-8B: Quality-control regenerator (`LLM_check.py`, Step 2)

```
You are a Medical Data Extractor. Convert the Report into a single structured line.

[CRITICAL RULES]:
1. **UNCERTAINTY RULE (AGGRESSIVE)**: If text says "potential", "possible", "
    ↪ suggesting", "likely" -> treat as **PRESENT**.
   - Example: "Potential necrotic areas" -> Necrosis: Present
   - Example: "Suggesting edema" -> Edema: Focal (or Extensive based on context)
2. **SIDE ASSIGNMENT**: If "Left frontal and parietal", write "Left frontal, Left
    ↪ parietal".
3. **FORMAT**: "Lesion: ... | Edema: ... | Necrosis: ... | Ventricles compression:
    ↪ ..."

[EXAMPLES]:
Input: "Hyperintense signal in left frontal and parietal lobes with potential
    ↪ necrosis."
Output: Lesion: Left frontal, Left parietal | Edema: Focal | Necrosis: Present |
    ↪ Ventricles compression: Absent

Input: "Right temporal tumor. Necrosis is visible and concentrated."
Output: Lesion: Right temporal | Edema: Focal | Necrosis: Present | Ventricles
    ↪ compression: Absent

Input: "Bilateral frontal lesions with significant edema."
Output: Lesion: Left frontal, Right frontal | Edema: Extensive | Necrosis: Absent |
    ↪ Ventricles compression: Absent

[CURRENT TASK]:
Input Report:
{src}

Specific Instruction (Fix this Error):
{error}

Output (Single line ONLY):
```

2.4 2d. Quality Control — Semantic Auditor Prompt

After syntax validation, the pipeline runs a semantic audit comparing the distilled text against the source report.

Listing 5: DeepSeek-R1-8B: Semantic auditor (`LLM_check.py`, Step 3)

```
Compare Source vs Distilled.
[Source]: {src}
[Distilled]: {dist}

Check for CRITICAL mismatches (Strict Logic):

1. **Necrosis/Edema Presence Rule**:
   - Source "Potential/Possible" vs Distilled "Present" -> **PASS** (Correct, we
    ↪ treat potential as present).
```

```
   - Source "Potential/Possible" vs Distilled "Absent" -> **FAIL** (Missed potential finding).
   - Source "Present" vs Distilled "Absent" -> **FAIL**.

2. **Edema Severity**:
   - Source "Significant/Extensive" vs Distilled "Focal" -> **FAIL**.
   - Source "Mild/Local" vs Distilled "Extensive" -> **FAIL**.

3. **Side Rule**:
   - Left vs Right Mismatch -> **FAIL**.

Ignore minor formatting or word order.

Output JSON: {"status": "PASS"} or {"status": "FAIL", "reason": "Short explanation"}
```

Quality-Control Pipeline Summary

The full quality-control pipeline (`LLM_check.py`) operates as a 5-step state machine:

1. **Syntax Check:** Verify 4-section structure, valid headers, no forbidden words (`lobe`, `bilateral`, `hyperintense`, signal/shape descriptors), valid laterality assignment, and whitelisted edema/ventricles values.
2. **Regenerate (if syntax fails):** Call the Generator prompt (Section 2.3) with the previous error injected via `{error}`.
3. **Semantic Audit (if syntax passes):** Call the Auditor prompt (Section 2.4) to verify critical attributes (necrosis/edema presence, severity, laterality) against the source report.
4. **Retry Loop:** Up to 5 iterations, each passing the accumulated error description to the regenerator.
5. **Save:** Overwrite the distilled file with the corrected version if fixes were applied.

The aggressive uncertainty policy treats all hedged statements ("potential," "possible," "suggesting") as affirmative for necrosis/edema presence to avoid missed findings.